Lecture 02: Milestones in Human Cultural and Economic Fronts of ICT

Connectivity: Empowering Development in Emerging Economies







Milestones in human, technological, cultural, and economic development can be attributed to the exchange and sharing of knowledge.

AGENDA

The World Today

Development Role of ICT

Operating Models

Vision of the Future

The World Today

The rate of technical innovation today is higher than ever before

Key innovations in manufacture, medicine, education

Techno-economic paradigm (Göransson & Söderberg, 2005):

- Innovations in technology (typically in comparatively developed countries) propel economic growth by increased productivity
  - 1700's Cotton & Iron production
  - 1800's Coal, Steam-engines, Railways
  - 1890's -1930's Steel, Vehicles, Chemicals, Telephony, Electricity
  - 1940's -1990's Oil, Material Synthesis, Aviation, Electronics
  - 1990's – Present: Microchip, Telecommunications, ICT

The pervasion of ICT & knowledge exchange drives rapid innovation & discovery in all fields; not simply 'IT'.

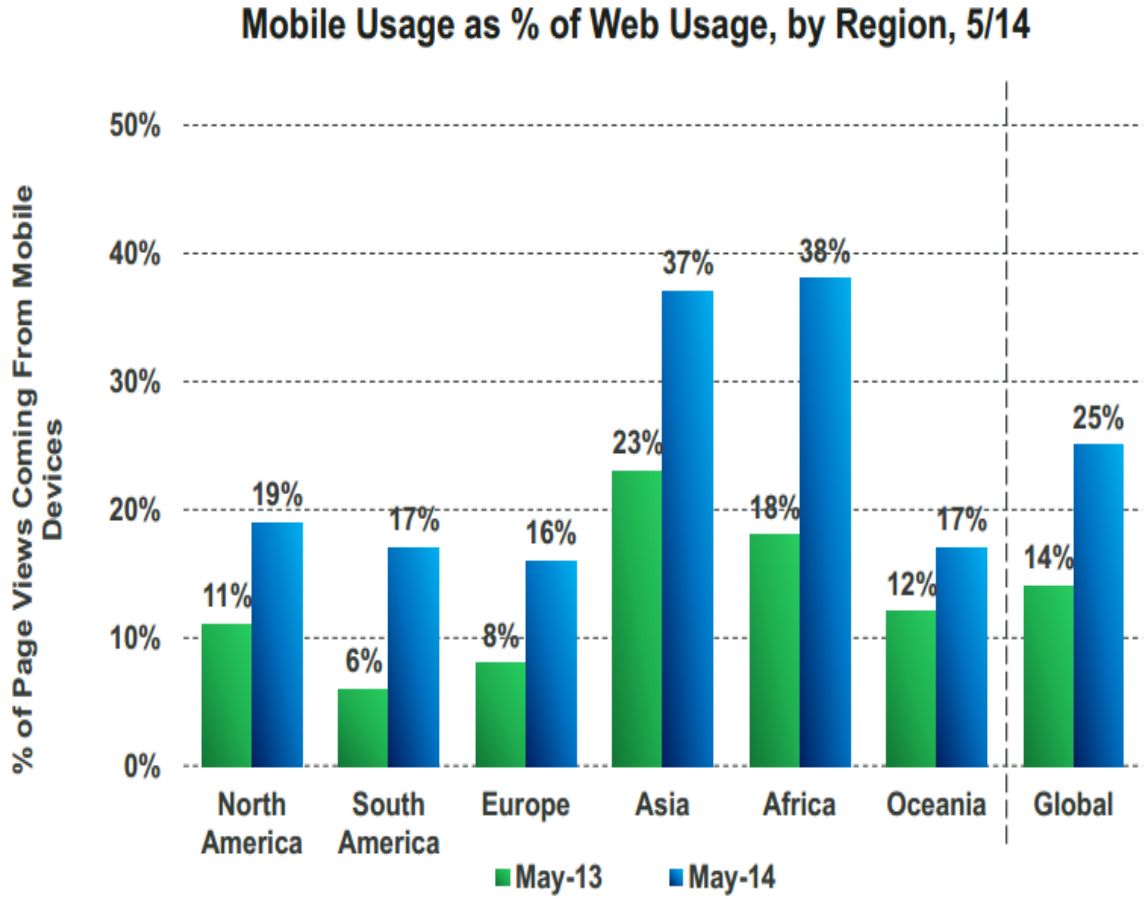
The state of today's tech: mobility, mobility and (unsurprisingly) more mobility.

~10% Year on Year uptake in internet penetration since 2013

20% growth in smartphone use in emerging markets: China/India/Brazil

52% increase in adoption of tablets

81% increase in mobile data traffic



Source: Meeker, M. 2014. *KPCB*

While trends suggest an optimistic future with promising internet growth and uptake in emerging economies...



Population: 7.18 bn

Internet Penetration: 39% (World)

- NA: 85%
- Asia: 32%
- Oceania* 67.5%
- Europe: 68.6%

Distribution of wealth:

- NA: 33.6%
- Africa: 1%
- Central America: 6.4%

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What role should ICT fulfil in defining an equitable, sustainable world?

The Development Role of ICT

What role can ICT play in the development of the world's economies?

ICT as a driver for change

- ICT as a resource must be adapted to a system
- It plays a catalytic role (Kelles-Viitanen, 2005)

An enabler – not a solution

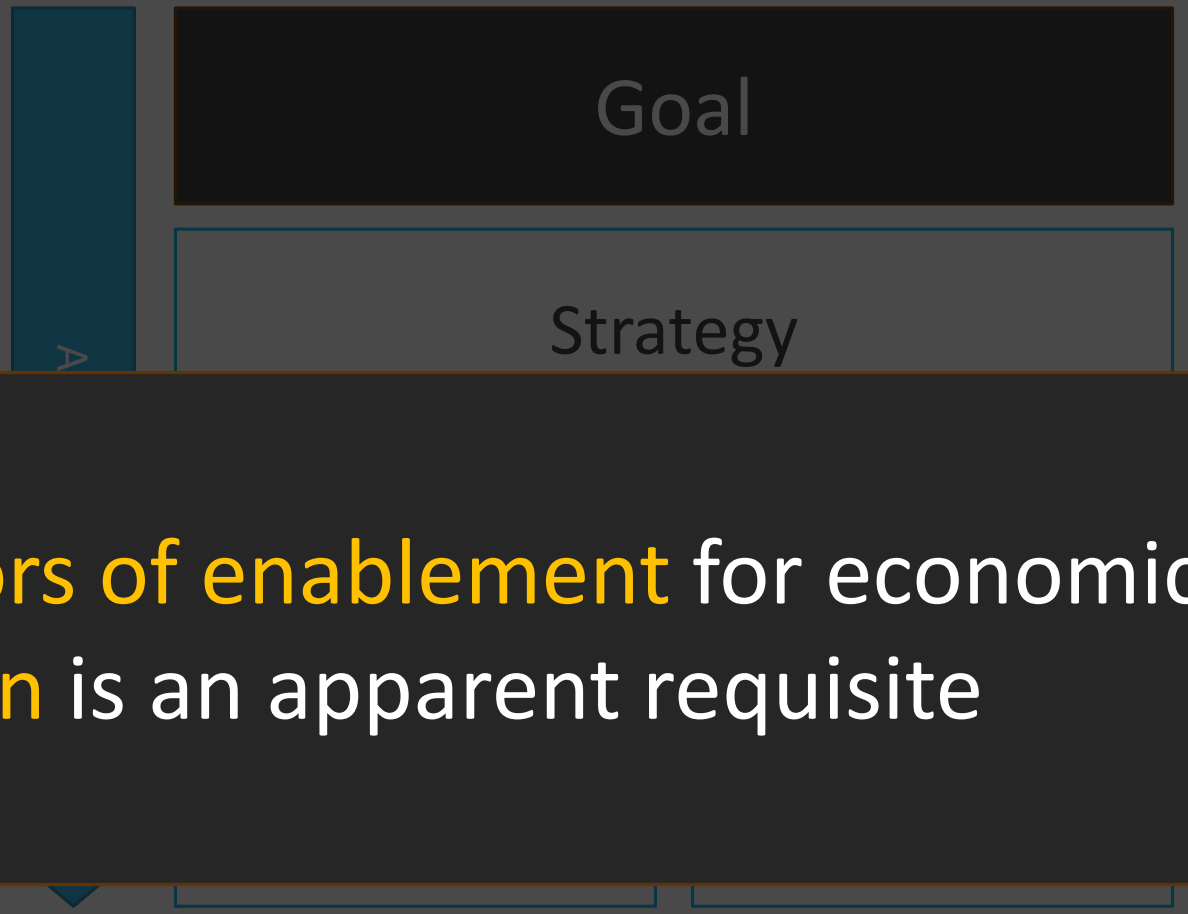


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While there are numerous factors of enablement for economic development, ICT adoption is an apparent requisite

Case 1: Commerce applications of ICT in rural implementations

While E-Commerce is relatively mature in developed economies, how has E-Commerce found its feet in the developing world?

- Challenges: Limited discretionary income
- Absence of delivery infrastructure
- Credential Challenges (raised by Ivan Fong)







(or technology enabled commerce) is transforming the way emerging economies do business

Knowledge transfer enabling more efficient value chains (Kenya, Morocco)

- Benefits to farmers; access to agricultural information, stock piles, sale prices, visibility over value chain (World Bank, 2013).
- Value chain brokers are publishing relay-information for transportation & logistics to mobile channels (World Trade Organisation, 2013).

Manobi Telecom (Senegal)

- Provides fishermen real-time weather & market pricing information via WAP & SMS.
- Fishermen can input fish stock
- Enabled the creation of an insurance policy which insured fishermen at sea

Case 2: Education & Human Development

ICT in education is pervasive.

Numerous attempts at bridging the connectivity gap

- USA: operating school busses outfitted with
- East Africa & South America: creating communally accessible computer-houses
- Malaysia: e-Melaka
- Africa, Asia & Oceania: OLPC

ICT literacy is recognised as a vital development goal; implementation less straight-forward.

Numerous attempts have been made to support ICT-literacy in education programmes in developing countries

Fixed labs

- Communal facilities. Fixed line connections, requires commute

Mobile labs

- Busses outfitted with computers & relevant equipment
- Driven to rural communities and schools
- More time-effective for communities, but expensive

PacRICS

- Pacific Regional Inter-Connectivity System

Project Loon

Case 3: Knowledge transfer in Agriculture, Healthcare, & Financial Services

Agriculture:

- Current market information establishes higher supplier-power
  - Higher margins
- Supply chain integration
- Lower transaction costs, publicise extension funds & other entitlement

Healthcare:

- Remote diagnosis of complicated diseases
- Administrative efficiencies

Financial Services:

- Mobile money. M-Pesa (Kenya), M-Paisa (Vodafone Fiji & BSP)

Challenges to deploying ICT (Hosman, 2012).

Electricity: inconsistent electricity provision

Cost: significantly lower discretionary income

Environmental Factors: humidity, salt, dirt, sunlight

Connectivity: poor (or non existent) connectivity

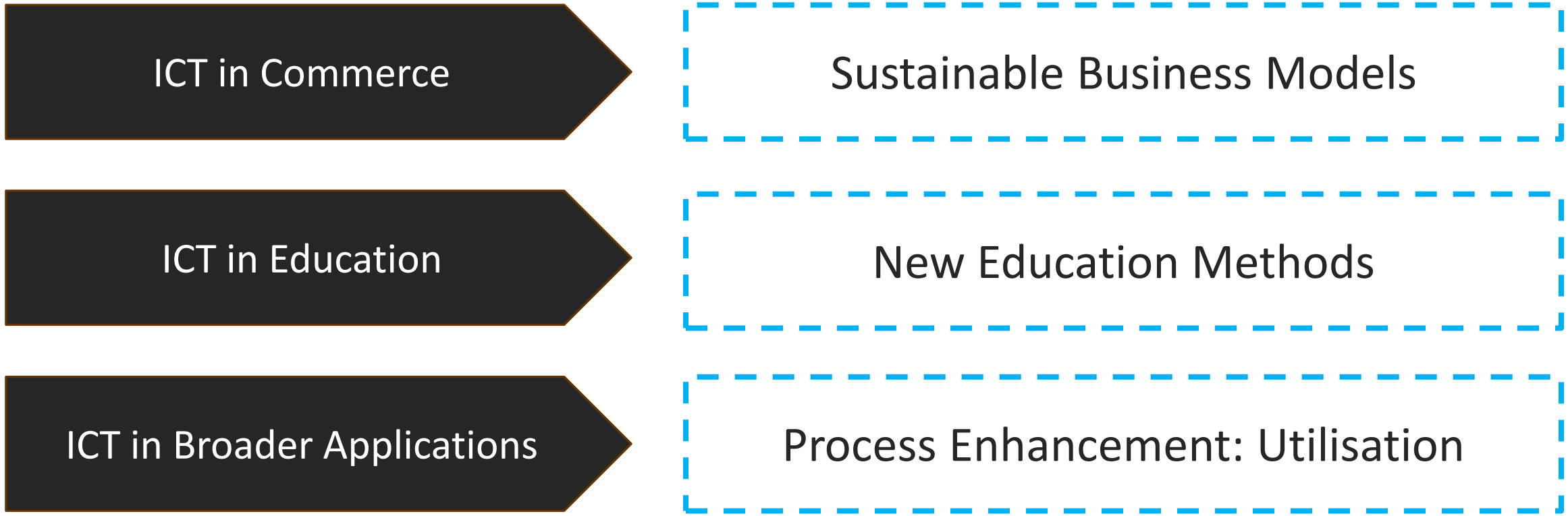
Maintenance & Support: limited ICT support available

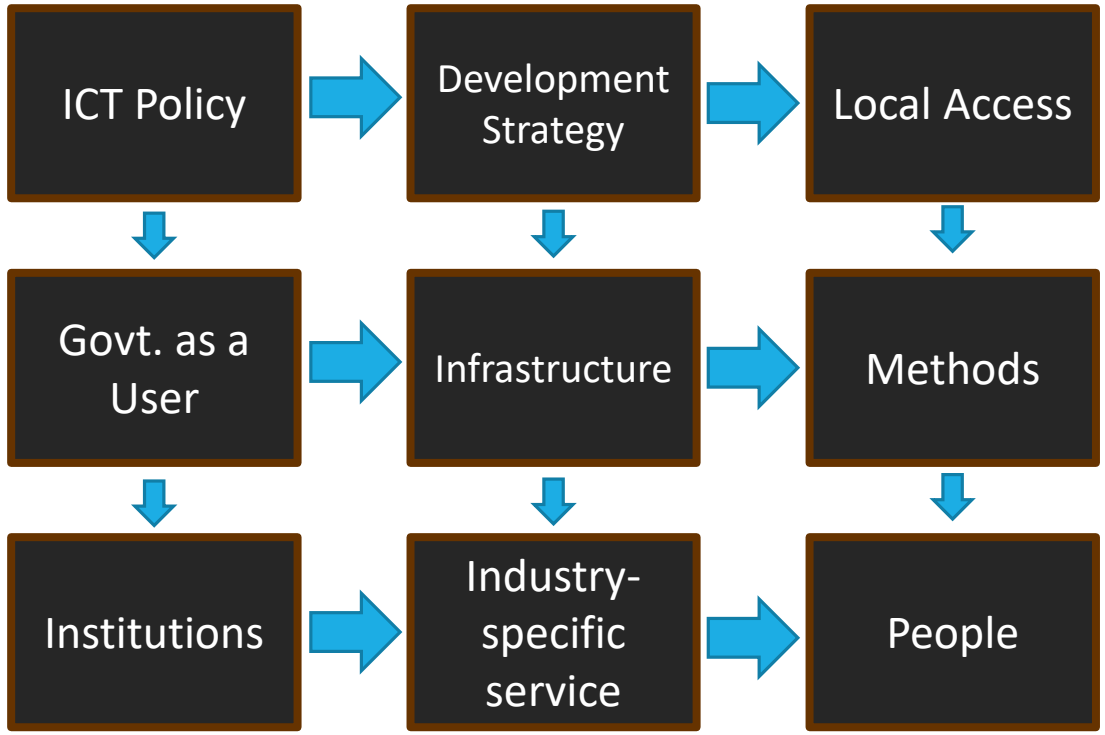
Other key considerations:

- Development initiatives should not be technology driven..

Adverse Operating Models

Having established the potential for change, how can ICT initiatives be operationalised?




ICT Policy
Development Strategy
Local Access
Govt. as a User
Infrastructure
Methods
Institutions
Industry-specific service
People

Open discussion: what strategies would you use to overcome the challenges in delivering ICT?

Challenges

1. Electricity
2. Cost
3. Environmental
4. Connectivity*
5. Maintenance

1. Application
    1. Commerce
    2. Education
    3. Healthcare / Agriculture
2. Medium
    1. TV Whitespace?
    2. Distributed Mesh?
    3. Satellite?
    4. Powerline Internet?

Thinking back to the key trends in the rest of the world, where's the confluence?

Trends

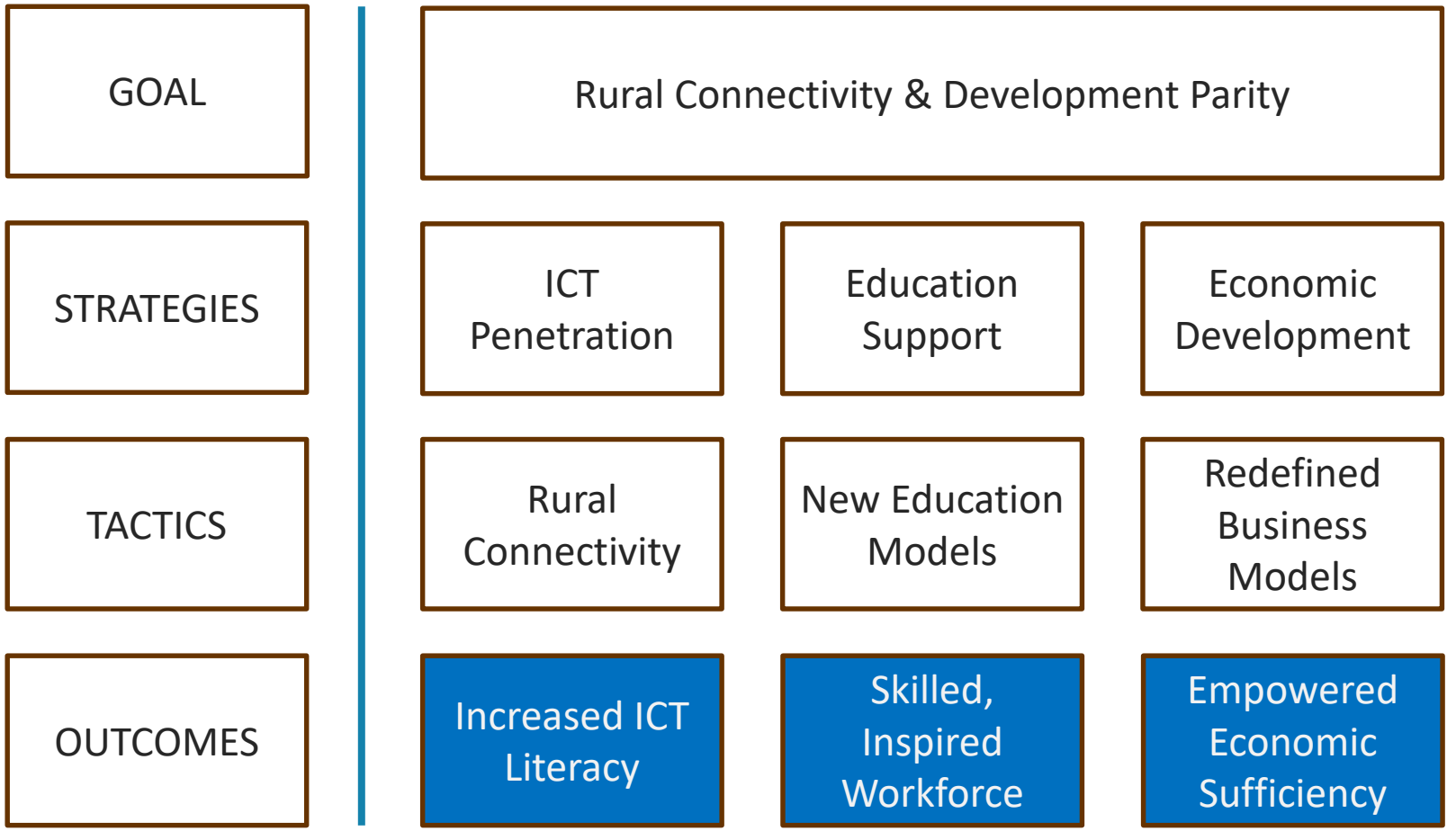
1. Increase in smartphone uptake
2. Growth in tablet market
3. Remarkable growth in mobile-based internet access

Rural Interconnectivity

1. Low-cost smart phones as primary ICT devices?
2. Capability of GSM/HSPA to support business models?
3. ? (Discussion outcomes)
4. ? (Discussion outcomes)

Vision of the Future

Vision of the Future



Move to Motion

APNIC	Internet Operations Community	Business Community	Student & Youth Community
ICT Policy Drive Support & focus on rural development	Methods and connectivity solutions for rural, widespread communities	New business models which take into account the comparatively lacking internet resources	Vision of how ICT can integrate with lifestyles, communities in a way that can sustainably enable growth

Thank You

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